# GUJARAT TECHNOLOGICAL UNIVERSITY
# DESIGN INNOVATION CENTRE

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# “Ergonomic Risk Assessment and Posture Monitoring using Wearable sensors”


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**Designation: Guide**

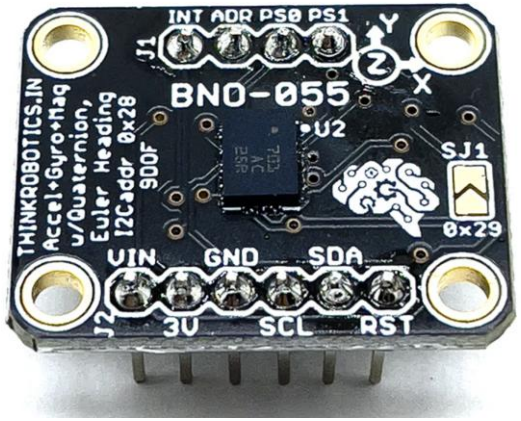
**Organization:GTU**

# Presentation Outline:

1. Introduction
2. Literature Review
3. Comparative Analysis/Gap
4. Problem Statement
5. Objective
6. Implementation Methodology
7. Work done/Results
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# Introduction:

- Construction workers frequently perform physically demanding tasks such as lifting, bending, and repetitive movements.
- These activities, when done incorrectly, can lead to serious injuries like back pain, muscle strain, and joint stress, affecting both worker health and productivity.
- This project aims to monitor workers' posture in real time using wearable motion sensors.
- These sensors track body movements such as trunk flexion, shoulder flexion, and elbow bending to detect unsafe postures.
- If a worker exceeds safe movement limits or maintains an improper posture for too long, the system provides immediate alerts, helping to prevent injuries before they occur.
- By implementing a wearable and portable system, this project contributes to workplace safety, injury prevention, and improved ergonomic practices in construction environments.




BNO055 9-DOF Absolute Orientation Sensor



Original ARDUINO NANO 33 BLE Board (Without Header) ABX00030 with built-in sensors LSM6D3

# Literature Review:

- Construction workers perform repetitive, physically demanding tasks that lead to musculoskeletal disorders (MSDs).
- Traditional methods like self-assessment surveys and observation-based analysis are prone to errors, time-consuming, and require expert supervision.
- Smartphone-based posture tracking has been explored but suffers from placement errors, limited accuracy, and lack of full-body motion tracking.
- IMUs (BNO055, LSM6DS3) provide high accuracy, real-time feedback, and complete motion tracking by measuring acceleration, rotation, and orientation.

**Existing Methods:**

**Smartphone Sensors (Accelerometer & Gyroscope)** – Placed on the **upper arm and waist** to track posture.

- **Issues**: Placement errors, limited tracking (only trunk & shoulder), no real-time feedback.

**Video-Based Observation** – Experts analyze posture from recorded videos.

- **Issues**: Time-consuming, requires professionals, not scalable for multiple workers.

**Why IMUs (Inertial Measurement Units)?**

- **High Accuracy** – IMUs (BNO055, LSM6DS3) track precise movements and angles, more accurate than smartphone sensors.
- **Real-Time Feedback** – Continuous monitoring enables instant posture correction, crucial for safety.
- **Complete Motion Tracking** – Measures acceleration, rotation, and magnetic field for full-body movement analysis.
- **Precise Body Placement** – Can be attached to specific body parts (trunk, shoulders, forearms) for better tracking.

# Comparative Analysis:

| SR No | Paper Title | Tool | Finding/Findings | Parameter |
|---|---|---|---|---|
| 01. | Wearable Mobile Sensors for Posture Monitoring in Construction Workers | • Smartphone Sensors | • Reliable for posture monitoring and risk detection, cost-effective, and easy to deploy. | • Trunk angles, Shoulder angles |
| 02. | IMUs for Dynamic Posture Tracking | • IMUs | • Superior accuracy and consistency for posture tracking but expensive and less scalable. | • Precision, Real-time adaptability |

| Feature | IMUs (BNO055, LSM6DS3) | Flex Sensors | Smartphone Sensors | Other (e.g., Pressure, Force) |
|---|---|---|---|---|
| Accuracy | High precision for motion and posture tracking | Limited accuracy for full-body motion | Varies; generally less accurate | Limited accuracy for posture tracking |
| Real-Time Feedback | Continuous real-time data and alerts | No real-time feedback | Limited real-time feedback | No real-time feedback |
| Motion Tracking | Tracks acceleration, rotation, and orientation | Tracks bending only | Basic acceleration and rotation | Tracks pressure or force, not motion |
| Posture Monitoring | Full-body posture tracking (flexion, rotation) | Best for joint angles | Basic posture and orientation | Not designed for posture tracking |
| Placement | Placed on specific body parts for accuracy | Typically on joints | On the smartphone | Depends on sensor type |
| Reliability | Consistent, minimal recalibration | Less reliable for full-body tracking | Varies by smartphone quality | Less reliable for dynamic tracking |
| Data Quality | High-quality, precise data | Affected by environmental factors | Varies, lower for posture tracking | Lower for detailed posture tracking |
| Integration | Easy with microcontrollers for real-time use | Complex setup and calibration | Easy, but limited for posture | Complex, not ideal for posture tracking |

# Identified Research Gap:

- **Limitations of Existing Research**
- Uses smartphone sensors, which are affected by phone placement and do not track full-body motion accurately.
- **Lacks real-time feedback** – postures are analyzed after data collection instead of being corrected instantly.
- Does not adapt to different body types, which affects posture accuracy for workers of varying heights and flexibility levels.
- **Not scalable for large worksites** – smartphones must be manually managed, limiting their use in multiple-worker monitoring.

- **How My Research Fills the Gap?**
- Uses IMUs for high-accuracy posture tracking instead of relying on smartphone sensors.
- Provides real-time alerts to correct posture instantly and prevent injuries.
- Integrates machine learning to adapt safe posture limits based on individual worker body types and flexibility.
- Scalable for large construction sites using wireless transmission and cloud-based monitoring.
- Multi-worker monitoring possible – A centralized dashboard allows tracking multiple workers at once for better safety enforcement.

# Problem Statement:

- Construction workers are at risk of developing musculoskeletal disorders due to poor posture over time. However, current systems lack real-time monitoring and corrective feedback, which makes it difficult to prevent these issues.
- Additionally, when scaling the solution to large construction sites, there is a need for a cloud-based dashboard to store and display posture data for multiple workers. A major challenge is that workers have different body structures, and posture angles vary depending on their height. To address this, machine learning algorithms are required to account for these individual differences and provide accurate posture assessments.
- This highlights the need for a precise, scalable, and reliable system that can monitor posture, give real-time feedback, and adjust for individual worker body variations

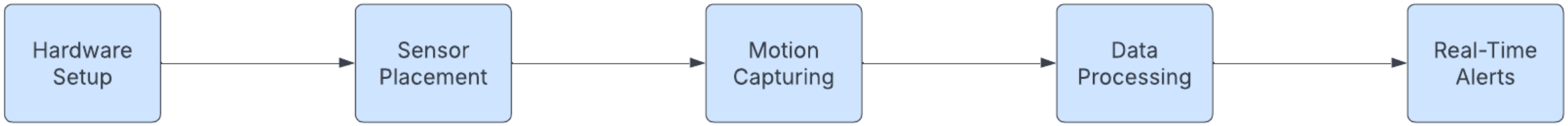
# Research Objective

- The goal of this project is to develop a real-time posture tracking system that uses wearable sensors, specifically IMUs, to monitor the posture of construction workers.
- By tracking their movements continuously, the system will help identify any poor posture, which can lead to musculoskeletal disorders over time. The system will analyze posture data to assess ergonomic risks and give detailed feedback to workers, helping them correct their posture instantly and prevent injuries.
- In addition to improving worker safety on construction sites, the system aims to provide scalable, cloud-based monitoring so that it can be easily used across large construction projects, tracking posture data for many workers at once
- The solution will also use machine learning to account for individual differences in body structure, such as height, ensuring that posture assessments are accurate for everyone.

# Implementation Methodology :

- **Wearable IMU-Based Posture Monitoring:** The system utilizes wearable IMUs (BNO055, LSM6DS3) to track worker posture in real-time by measuring movements such as bending, twisting, and rotating.
- **Wireless Data Transmission & Calibration:** The collected sensor data is transmitted wirelessly via Bluetooth (BLE) or Wi-Fi to a microcontroller (Arduino or Raspberry Pi) for real-time processing.
- **Data Processing:** The collected data is processed to identify risky posture angles that could lead to musculoskeletal disorders. The system analyzes posture to track the worker's movements and spot any angles that fall within a "risky" range.
- **Risk Duration:** The system also measures how long the worker holds a risky posture. If the worker maintains a harmful position for too long, the system flags it as a potential ergonomic risk.
- **Real-Time Feedback:** The system provides instant feedback (such as alerts or notifications) to let the worker know when they are in a risky posture, helping them correct it before it causes injury.
- **Posture Data Accuracy:** The system ensures the accuracy of posture data by cross-checking readings from multiple sensors placed on different body parts, reducing errors and improving the reliability of the risk assessment

# Proposed WorkFlow




Proposed Workflow Diagram

# Work Done :

### Hardware & Scalability Analysis
- Focused on Arduino Nano 33 BLE for real-time posture tracking.
- Studied power requirements and efficiency for wearable sensor deployment.

### Component Understanding & Data Processing
- Deep dive into IMUs (BNO055, LSM6DS3) for sensor calibration & data fusion.
- Explored flex sensors (optional) and their relevance to posture tracking.
- Researched real-time alerts (beep sounds, Bluetooth notifications, cloud alerts).

### Software & Communication Protocols
- Reviewed communication protocols (I2C for IMUs, Analog for flex sensors).
- Studied how data processing determines risky posture angles and duration.

### Project Documentation & Presentation
- Worked on PPT preparation for project updates.
- Finalized hardware components (IMUs, microcontrollers, power source).

### Dashboard Research & Cloud Integration
- Studied ThingsBoard for real-time dashboards and data visualization.
- Understood how ThingsBoard stores and compares sensor data over time.

# Future Work :

**Hardware Setup & Integration**

- Connect and manage IMUs, Arduino Nano 33 BLE, battery, and jumper wires for real-time posture tracking.

**Data Collection & Processing**

- Fetch raw sensor data from IMUs and convert it into CSV/Excel format for angle analysis.

**Real-time Dashboard Development**

- Build a dashboard to visualize posture data and identify risky postures for large-scale construction sites.

**Posture Risk Classification**

- Analyze sensor data to determine safe and unsafe posture thresholds.

**Machine Learning Integration**

- Train a model to classify posture risks automatically based on sensor input.

**Final Product Deployment**

- Ensure real-time monitoring, alerts, and scalability for construction industries.

# THANK YOU!!